



Spectrum annotation with transformer models

Attila Kertesz-Farkas, Nina Krivulets

ФКН НИУ ВШЭ, Москва

Core concepts

Mass spectrometry is an analytical technique used to measure the mass-to-charge ratio (m/z) of ions. It helps identify the chemical composition of a sample and determine the structure of molecules. It is widely used in chemistry, biology, and related fields for analyzing small molecules, proteins, metabolites, and other substances.

Mass spectrum: A graphical representation showing the intensity of ions at various m/z ratios.

Peaks: Each peak in the mass spectrum represents an ion with a specific m/z ratio, corresponding to a molecular fragment or the intact molecule, and heights of the peaks connote the relative abundance of the various components in the sample.

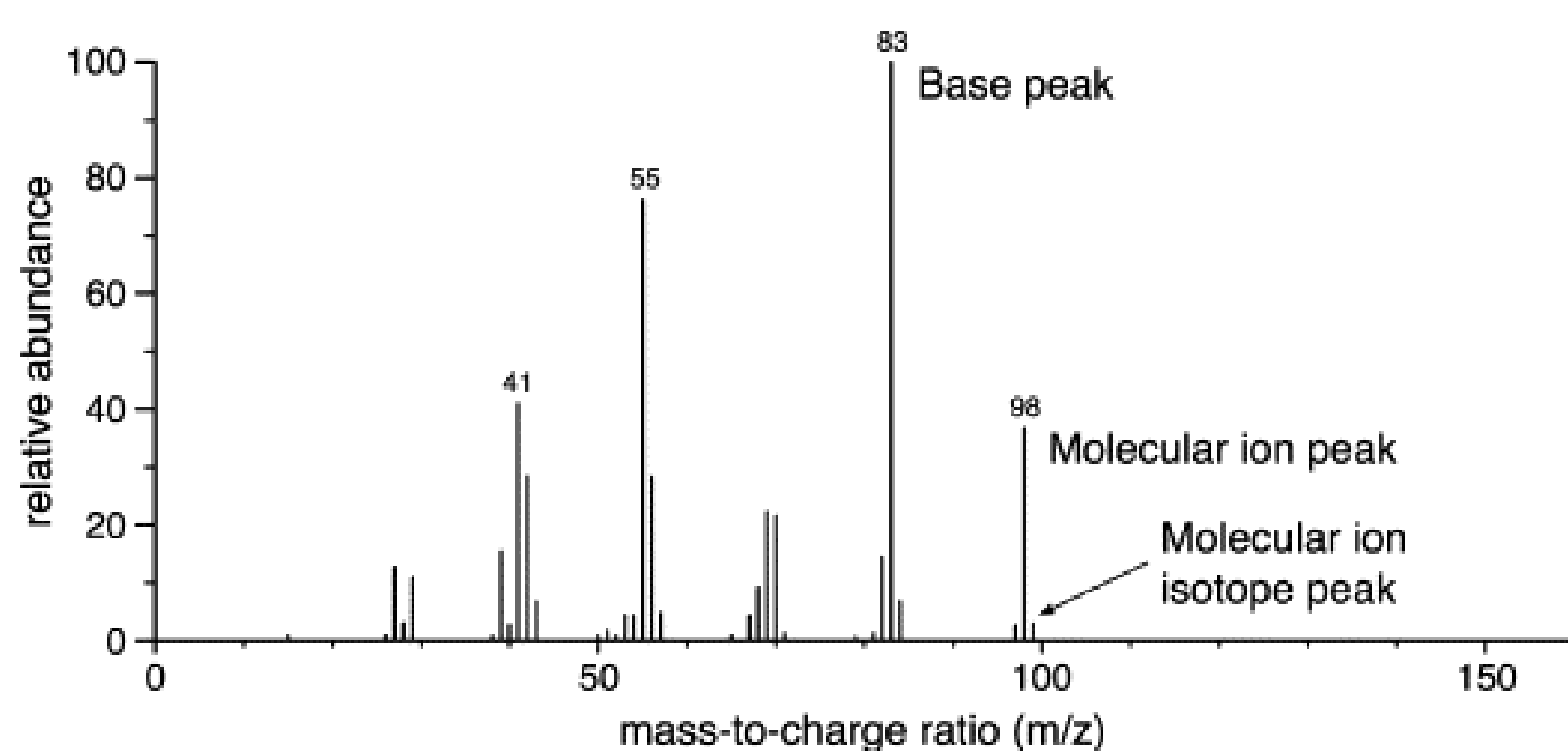


Рис. 1: Mass spectrum

Architecture overview

In this study, transformer models are used for deep learning. These models were initially introduced in the paper [1].

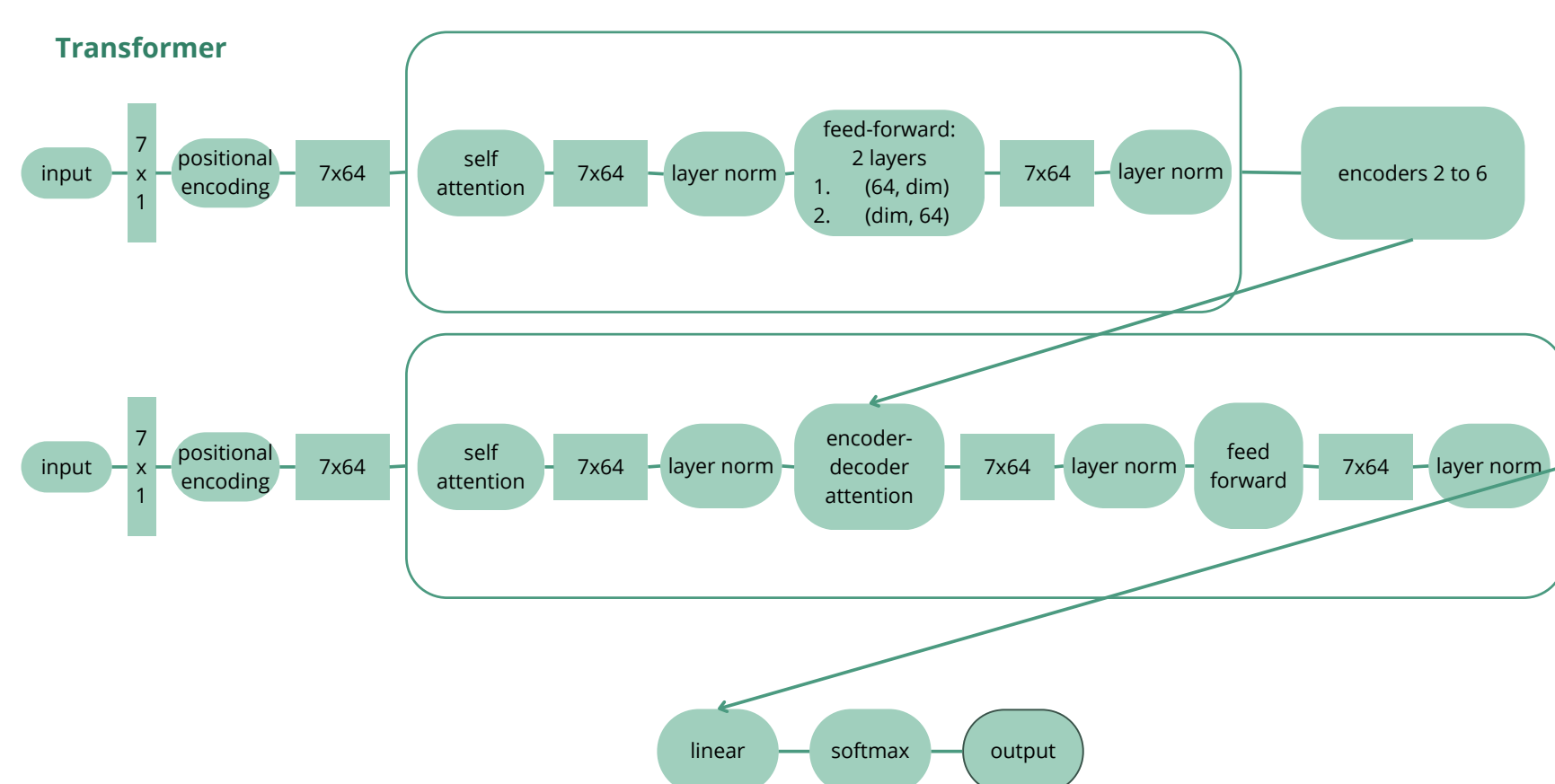


Рис. 2: Transformer architecture

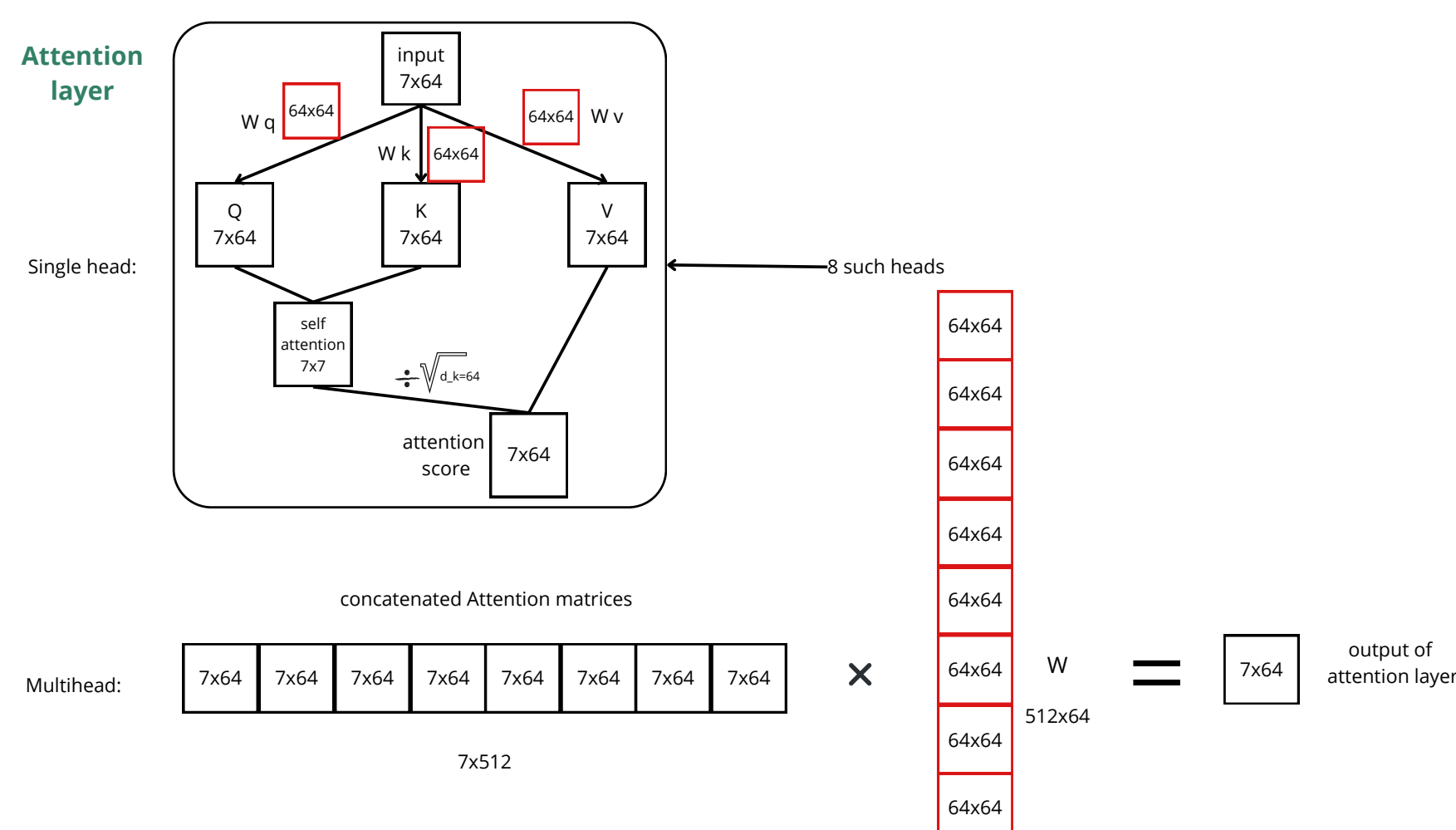


Рис. 3: Attention layer

The Transformer architecture, introduced in the paper "Attention Is All You Need", revolutionized deep learning by eliminating the need for recurrent or convolutional layers. Instead, it relies entirely on self-attention mechanisms to process input data, enabling parallelization and faster training.

The architecture consists of an encoder and a decoder, both made up of multiple layers. Each encoder layer has two main components: multi-head self-attention and position-wise feedforward networks. The self-attention mechanism allows the model to weigh the importance of different input tokens relative to each other, enabling it to capture dependencies regardless of distance. Multi-head attention enhances this by allowing the model to focus on different aspects of the input.

The decoder, similarly structured, includes additional attention layers that focus on the encoder's output, enabling sequence-to-sequence tasks. Positional encoding is added to preserve the order of tokens in sequences, as the architecture lacks inherent recurrence.

Current progress

For now, we were able to verify the data, as one of the problems in mass spectrometry is that 60-80% of the experimental spectra cannot be annotated correctly with high confidence.

Positional encoding based on the article [1] has been implemented for the data. Formula:

$$PE_{(pos,2i)} = \sin\left(\frac{p}{10000^{\frac{2i}{d_{model}}}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{p}{10000^{\frac{2i}{d_{model}}}}\right)$$

where p is the peak value, i is the position and d_{model} is the dimension of the model.

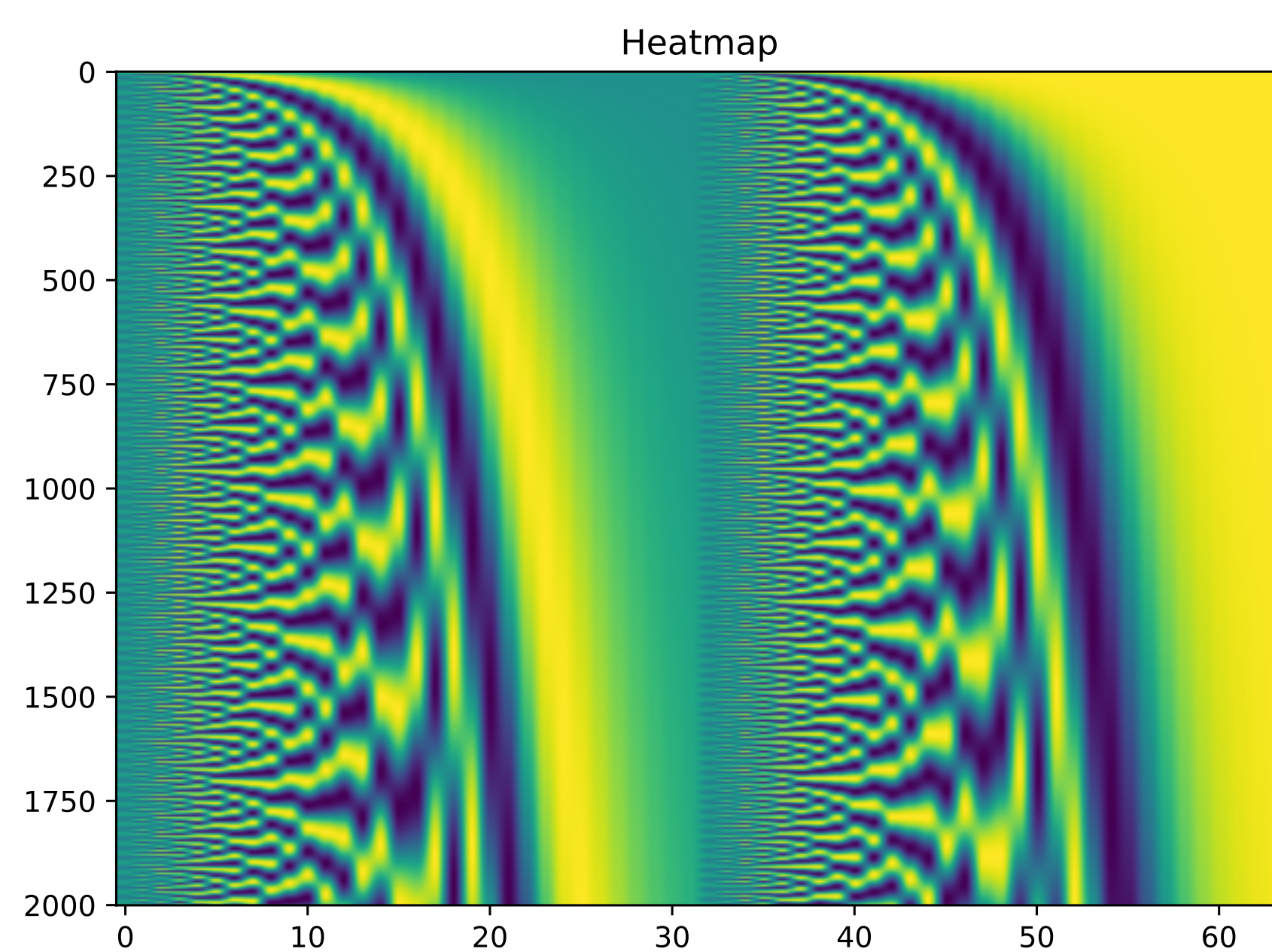


Рис. 4: Heatmap of the positional encodings

Future research

The transformer neural network is currently being implemented, and after utilizing it, tests need to be conducted and the model will have to be fine-tuned. To evaluate the model's performance, it will also be compared with existing solutions.

References

- [1] Ashish Vaswani and Noam Shazeer and Niki Parmar and Jakob Uszkoreit and Llion Jones and Aidan N. Gomez and Lukasz Kaiser and Illia Polosukhin. "Attention Is All You Need". arXiv:1706.03762.
- [2] Jimmy Lei Ba and Jamie Ryan Kiros and Geoffrey E. Hinton. "Layer Normalization". arXiv:1607.06450.